

HELLO!

I am Sayeed Unnisa

I am here because I love to give presentations on **Java Full Stack**

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SPRING BOOT REST (WEBSERVICES)



WebServices

- Integration of two different applications which might be running in same Server or different Server, also applications are developed in same language/different language
- Integration of two or more applications to work together (for communication purpose) we use Webservices.
- Linking two or more applications using HTTP protocol and Global Data Format (XML/JSON)
- Minimum two applications are required, in that one application behaves like Provider and another application behaves like Consumer.

Rest Web Services (ReSTful Web Services)

- ReST stands for Representational State Transfer.
- Representation means “Global Format”
- State means “Data”
- Transfer means “sending (request) and receiving (response)”.

Here Global Formats are XML, JSON.

- XML ---> Extensible Markup Language
- JSON ---> Javascript Object Notation.

Rest Web Services (ReSTful Web Services)

- In Rest web services two applications data (object) will be converted into XML/JSON Format, again converted back to language supported format.
- XML : eXtensible Markup Language : This is used to represent data which can be understood by almost every high level language.
Ex: Java, .net, PHP, Python...etc
- JAXB [Java Architecture for XML Binding] API is used to convert
Object-> XML (Marshalling) and
XML -> Object (Unmarshalling)
- JSON (Java Script Object Notation) is light weight compared to XML and faster in conversions, transfer data over network.
- **Syntax of JSON:**
- { "key" : value, }

Note: keep value in quotes if String type

HTTP: Hyper Text Transfer Protocol

- Http Contains 2 types of Messages:
 1. Request : Consumer to Producer (ask todo some work)
 2. Response : Producer to Consumer (work done/fail and its status)

- Either Request or Response contains two parts Head and Body.
 - Head contains initial line and Header Params (K:V)
 - Body contains Data(Text/HTML/XML/JSON/..etc)

➤ For Request Http method must be specified.

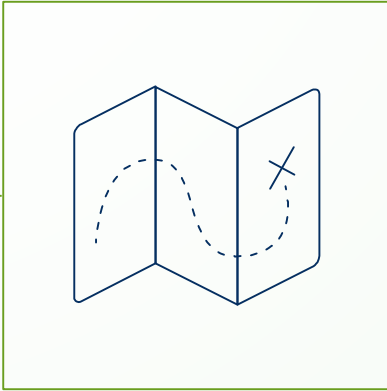
1. GET : Fetch Resource from producer
2. POST: Create new Resource at producer
3. PUT : Modify existing Resource at producer
4. DELETE : Remove existing Resource at producer
5. PATCH: Partially modify existed Resource at producer

➤ Http Response contains status code that says what happened at Producer while processing request.

Code	message type
1xx	Information
2xx	Success
3xx	Redirect
4xx	Client Side Error
5xx	Server side Error

- Webservices Code is written in 4th Layer i.e. Integration Layer.
- This layer must exist at both Producer and Consumer.
- Consumer contains code at Integration Layer i.e. called as Stub
- A code that makes http request: Stub
- Producer contains code at Integration Layer i.e. called as Skeleton
- A code that process request and Gives http response: Skeleton

- Spring Boot Rest Producer application can be implemented using RestController Concept which returns final output as : `Rtity<T>`
`@RestController = @Controller + @ResponseBody`
- Here we need to define producer/consumer application using Spring web dependency only based on Annotations `@Controller` and `@RestController`
- `ResponseEntity<T>` contains 3 things mainly
 - Response Body (can be null),
 - Response Headers (can be null)
 - `HttpStatus` (must be given)



THANKS!

Any questions?

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